

Euclidean and Non-Euclidean Geometry:

Exercises:

1. a) Negate $P \vee Q = \sim(P \vee Q)$
 $= P \nrightarrow Q$.. when exclusive

b) Negate $P \& \sim Q = \sim(P \& \sim Q)$
 $= \sim P \vee Q$

c) (pg 60) "Logic Rule #3"
 $\sim(\sim S) = S$

(pg 61) "Logic Rule #4"
 $\sim[H \Rightarrow C] = H \& \sim C$

(pg 61) "Logic Rule #5"
 $\sim[S_1 \& S_2] = \sim S_1 \vee \sim S_2$

$P \Rightarrow Q$ is $\sim(P \& \sim Q)$

which also $\sim P \vee Q$

2. (pg 19) "Euclid's Fourth Postulate"

"All right angles are congruent to one another"

Negation to Euclid's Fourth Postulate:

No right angles are congruent to many other.

3. (pg 21) "Euclid's Parallel Postulate"

"For every line l and for every point P that does not lie on l , there exists a unique line m through P that is parallel to l "

Negation to Euclid's Parallel Postulate:

"For a line l and for a point P not on l , there exists a unique line m through P not parallel to l ."

4. (from exercise) "

"If lines l and m are parallel, then a transversal t to lines l and m cuts out congruent alternate interior angles."

Converse:

"Traversal t to line l and m cuts congruent alternate interior angles, if lines l and m are parallel."

b) (from exercise)

"If the sum of the degree measures of the interior angles on one side of transversal t is less than 180° , then lines l and m meet on that side of transversal t ."

Converse:

"Lines l and m meet at a transversal t , if the sum of the degree measures of the interior angles on one side of t is less than 180° ."

5.

(pg 71) "Proposition 2.2"

"There exist three distinct lines that are not concurrent."

Proof:	<u>Hypothesis</u>	<u>Evidence</u>
	1) A line l and m exist	Euclid's Parallel Postulate
	2) A third line n is possible not parallel to l and m .	Proposition 2.1

6.

(pg 71) "Proposition 2.3"

"For every line, there is at least one point not lying on it."

Proof:	<u>Hypothesis</u>	<u>Evidence</u>
	1) Three points exist; $A, B,$ and C	(pg 12) "Definition-Point"
	2) A line l only passes through two points maximum	Incidence Axiom # 3.

(pg 71) "Proposition 2.4"

"For every point, there is at least one line not passing through it."

Proof:	Hypothesis	Evidence
	1) Three points exist; A, B, and C	(pg 12) "Definition-Point"
	2) Two points form a line, and third point alone	Incidence Axiom #3

(pg 71) "Proposition 2.5"

"For every point P, there exist at least two distinct lines through P."

Proof:	Hypothesis	Evidence
	1) Lines l and m pass through point P , but are not parallel	Proposition 2.1

7

(pg 69) "Incidence Axiom - 1"

"For every point P and for every point Q not equal to P, there exists a unique line l incident with P and Q."

$$\boxed{\forall P \forall Q ((P \neq Q) \Rightarrow \exists ! l (P I l \ \& \ Q I l))}$$

(pg 69) "Incidence Axiom - 2"

"For every line l, there exists at least two distinct points incident with l"

$$\boxed{\exists P \exists Q (P \neq Q \ \& \ (P I l \ \& \ Q I l))}$$

(pg 69) "Incidence Axiom - 3"

"There exist three distinct points with the property that no line is incident with all three of them"

$$\boxed{\exists A \exists B \exists C ((A \neq B \ \& \ A \neq C \ \& \ B \neq C) \ \& \ \sim \exists l (A I l \ \& \ B I l \ \& \ C I l))}$$

Axioms		Visual	Notation
Dependent	Independent		
1, 2	3		$\exists I \Rightarrow \exists P \exists Q ((P \neq Q) \& (P I Q \& Q I P))$
1, 3	2		$\forall P \forall Q \forall R ((P \neq Q \& Q \neq R) \Rightarrow \exists I (P \& Q \& R I I))$
2, 3	1		$\forall I \exists P \exists Q \exists R (P \neq Q \& Q \neq R \& \exists I (P \& Q \& R I I))$

8.

(pg 74) "Example 3"

$ab \in \{A, B\}$, $ac \in \{A, C\}$, $ad \in \{A, D\}$, $bc \in \{B, C\}$, $bd \in \{B, D\}$
and $cd \in \{C, D\}$

(pg 74) "Example 4"

$\exists A \exists B \exists C \exists D \exists E$

$ab \in \{A, B\}$

$ac \in \{A, C\}$

$ad \in \{A, D\}$

$ae \in \{A, E\}$

$bc \in \{B, C\}$

$bd \in \{B, D\}$

$be \in \{B, E\}$

$cd \in \{C, D\}$

$ce \in \{C, E\}$

$de \in \{D, E\}$

Incidence Geometry:

Axiom 1 matches Examples #1 and #2

When $P = A|B|C|D|E$ & $Q = A|B|C|D|E$ & $P \neq Q$

Axiom 2 interpretes well if

$$l = ab|ac|ad|ae|bc|bd|be|cd|ce|de$$

$$\text{because } l \perp \{\{A,B\}|\{A,C\}|\{A,D\}|\{A,E\}|\{B,C\}|\{B,D\}|\{B,E\}|\{C,D\}|\{C,E\}|\{D,E\}\}$$

Axiom 3 correlates when $P = \{A|B|C|D|E\}$

$$\& Q = \{A|B|C|D|E\} \& R = \{A|B|C|P|E\}$$

$$\& P \neq Q \& P \neq R \& Q \neq R \Rightarrow l = \{\{AB\}|\{AC\}|\{AD\}|\{AE\}|\{BC\}|\{BD\}|\{BE\}|\{CD\}|\{CE\}|\{DE\}\}$$

$$\{AD\}|\{AE\}|\{BC\}|\{BD\}|\{BE\}|\{CD\}|\{CE\}|\{DE\}\}$$

Euclidean Parallel Property:

For every line $l = \{\{AB\}|\{AC\}|\{AD\}|\{AE\}|\{BC\}$

$|\{BD\}|\{BE\}|\{CD\}|\{CE\}|\{DE\}\}$ and for

every point $P = \{A|B|C|D|E\}$ not on

line l , there exists a unique line

$m = \{\{AB\}|\{AC\}|\{AD\}|\{AE\}|\{BC\}|\{BD\}|\{BE\}$

$|\{CD\}|\{CE\}|\{DE\}\} \& m \perp P \& m \perp l$.

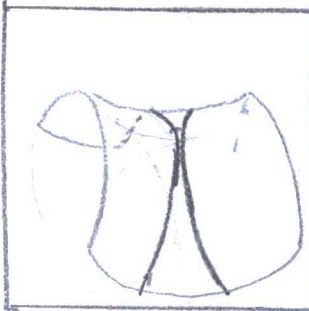
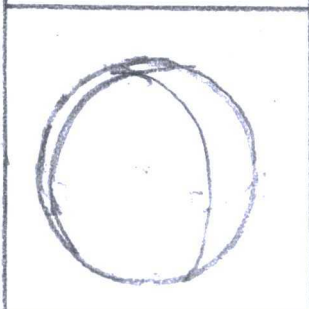
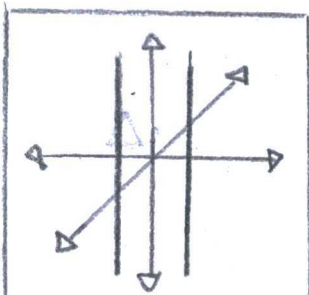
Hyperbolic Parallel Property:

For every line $l = \{\{A, B\} | \{A, C\} | \{A, D\} | \{A, E\} | \{B, C\} | \{B, D\} | \{B, E\} | \{C, D\} | \{C, E\} | \{D, E\}\}$ and every point $P = \{A | B | C | D | E\}$ not on l , there exists at least two lines through P parallel to l .

Note: Only when P is the reference point.

9.

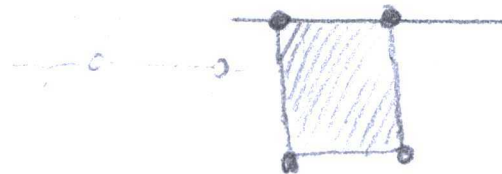
a) Points $\rightarrow$ Lines



Lines $\rightarrow$ Planes



Incidence is a line on a plane



"Euclidean,
Elliptic,
or Hyperbolic" lines"

Incidence Axiom			Euclidean	Elliptic	Hyperbolic
1	2	3			
Yes	Yes	Yes	Yes	No	No

b)

Incidence Axiom			Euclidean	Elliptic	Hyperbolic
1	2	3			
Yes	Yes	Yes	Yes	No	No

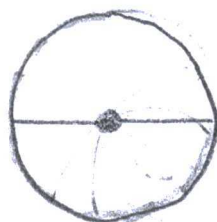
c) Point $\longrightarrow$ Point in circle



Line $\longrightarrow$ Chord of a circle

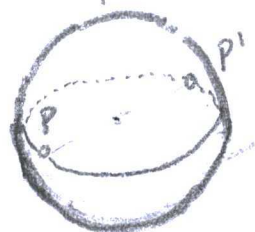


Incidence means a point on a chord



Incidence Axiom			Euclidean	Elliptic	Hyperbolic
1	2	3			
Yes	Yes	No	Yes	No	No

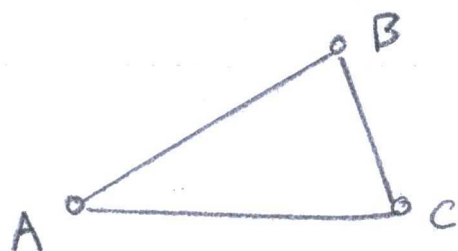
d) Antipodal points on a sphere



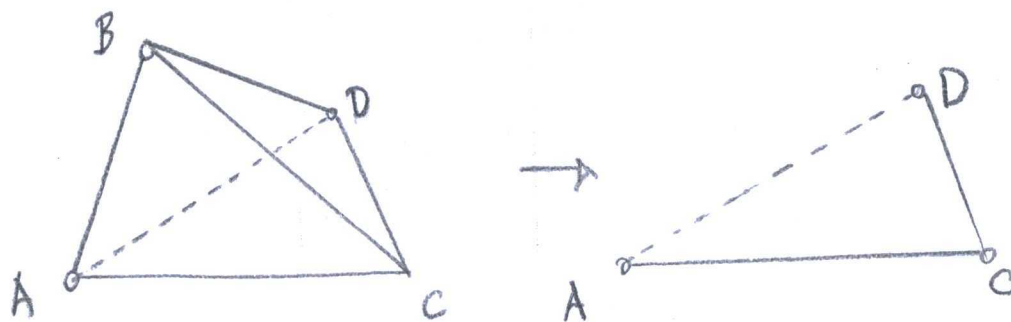
Incidence Axiom			Euclidean	Elliptic	Hyperbolic
1	2	3			
Yes	Yes	No	Yes	Yes	No

10.

a) A model about three point incidence geometry is in Figure 2.4. The figure below shows a three point model.

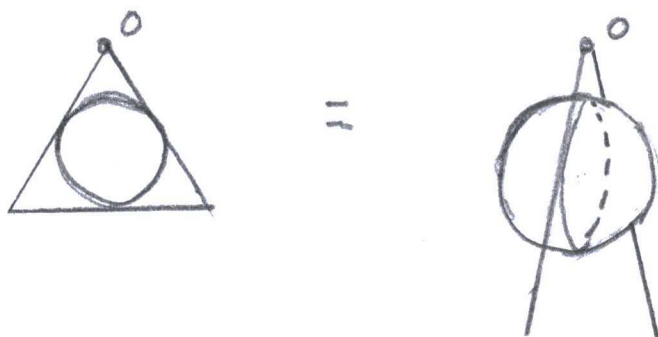


The other model has all four points in Figure 2.5, but "isomorphic" after a reduction to three-point.

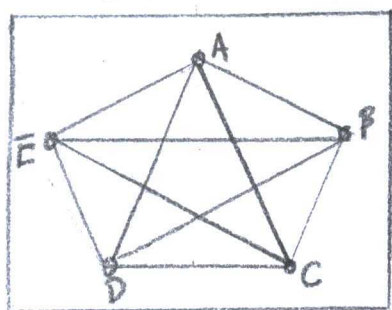


b) The models are isomorphic with three or four points.

c) Models in 9(b) and 9(d) are isomorphic.



11.



"Neither elliptic, hyperbolic, nor Euclidean parallel properties."

(pg 82) "Elliptic Parallel Postulate"
"any two lines meet"

(pg 21) "Euclidean Parallel Postulate"
"For every line l and for every point P that does not lie on l , there exists a unique line m through P that is parallel to l "

(pg 75) "Hyperbolic Parallel Postulate"
"For every line l and every point

P not on l , there exist at least two lines through P parallel to l "

The image has no parallel lines in geometries above.

12.

a) (from exercise) "Transitive Parallelism"

$$\forall l \forall m \forall n (l \parallel m \ \& \ m \parallel n \ \& \ l \neq n \Rightarrow l \parallel n)$$

(pg 82) "Affine Plane"

$$\forall l \forall P (\sim(P \in l) \Rightarrow \exists ! m (P \in m \ \& \ l \parallel m))$$

Similarly,

$$\forall l \forall P (\sim(P \in l) \Rightarrow \exists ! m (P \in m \ \& \ l \parallel m))$$

$$\ \& \ (P \in m) \Rightarrow \exists ! n (P \in n \ \& \ m \parallel n)$$

$$\ \& \ l \neq n)$$

b) The notation $l \neq n$ was important because unique lines ($\exists !$) and relationship between the lines.

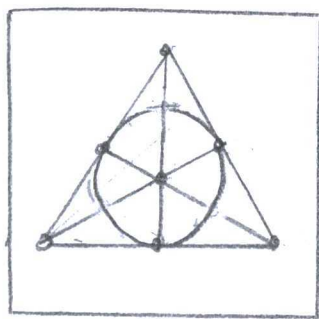
c) Converse model°

$$\forall l \forall m \forall n (\exists !_m (p I m \ \& \ l \parallel m) \Rightarrow \sim (p I l))$$

$$\& \exists !_n (p I n \ \& \ m \parallel n) \Rightarrow \sim (p I m) \\ \& \ l \neq n)$$

d) Parallel lines are not transitive in part a
with no $l \neq n$.

13.



Fano's plane is a model with
seven lines and points. Three
intersections make a point. with
three points on every line.

Higher-order finite projections are
possible, but hard on paper.

14.

a) (from exercise) "Statement S"

"If l and m are any two distinct lines,
then there exists a point P that
does not lie on either l or m "

(pg 69) "Incidence Axiom #1"

$$\forall P \forall Q ((P \neq Q \Rightarrow \exists ! l (P \in l \& Q \in l)) \Rightarrow \\ \exists ! m (\neg (R \in m \& R \in l)))$$

True to false implication is false.

(pg 69) "Incidence Axiom #2"

$$\forall l \exists P \exists Q ((P \neq Q) \& (P \in l \& Q \in l)) \Rightarrow$$

$$\forall R ((R \neq P \& R \neq Q) \Rightarrow \exists ! m (\neg (R \in m \& R \in l)))$$

True implication to false is false.

(pg 69) "Incidence Axiom #3"

$$\exists A \exists B \exists C ((A \neq B \& A \neq C \& B \neq C)$$

$$\& \sim \exists l (A \in l \& B \in l \& C \in l))$$

$$\Rightarrow \exists R ((R \neq A \& R \neq B \& R \neq C) \Rightarrow$$

$$\exists ! m (\neg (R \in l \& R \in m)))$$

b) (pg 82) "Projective Plane"

Requirements: (1) Elliptic Parallel Property

(2) Incidence Axiom #2

As an example, the logic above justified proof from incidence axioms, specifically, "statement S is not proved by incidence axioms." So, the negate statement $\sim S$ is not provable either.

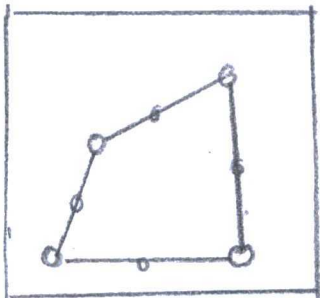
c) All lines have the same number of points lying on them:

Proof:	<u>Hypothesis</u>	<u>Evidence</u>
	1) Two lines exist, l and m , along with a point P not on l or m .	Statement S
	2) Another point Q forms line PQ	Euclid's Postulate I
	3) PQ is parallel to l and m or not	Euclid Postulate V

d) All lines have the same number of points lying on them.

Proof:	<u>Hypothesis</u>	<u>Evidence</u>
	1) Two lines exist, l and m , also point P not on l or m .	Statement S
	2) A point Q forms line PQ	Euclid Postulate I
	3) PQ is parallel to l and m , or not	Euclid Postulate IV

15



"Quadrangle"

a) A quadrangle exists when every line has at least three points lying on it:

Proof:	<u>Hypothesis</u>	<u>Evidence</u>
	1) Two projective planes P and Q	(Pg 32) "Projective Planes"

have two lines
m and o, also u
and v.

2) Lines intersect
into a quadrangle
and have three
points

Axiom I-3.

c) A proposition when quadrangles
never appear in affine planes.

$$Z = y \quad Z_0 = y + 1$$

$$Z_0 = X \quad Z_1 = X - 1$$

16.

a) (from Example 8) "line"

$$ax + by + c = 0$$

(pg 87) "linear homogenous equation"

$$ax + bx + cz = 0$$

Verify Axiom I-1:

Distinct points: (u, v) and (s, t)

$$\text{2D-plane: } ax+by+c=as+bt+c=0$$

$$\text{3D-space: } ax+by+cZ=au+bv+cZ=0$$

$$a(u-s)+b(v-t)+c(Z-1)=0$$

implies a single point

shifted.

Verify Axiom I-2:

$$\text{If } a \neq 0, \text{ then } (x, y) = \left(-\frac{c}{a}, 0\right)$$

$$\text{Else } b \neq 0, \text{ then } (x, y) = \left(0, -\frac{c}{b}\right)$$

... again a single point on a line.

Verify Axiom I-3:

With three points $(0, 0)$, $(0, 1)$,

and $(1, 0)$, the linear equation

is not collinear.

$$a(0)+b(0)+c=a(0)+b(1)+c$$

$$=a(1)+b(0)+c$$

$$=0, a=b=c=0$$

Verify Euclidean Parallel Property:

Parallel if slopes equal by points
Such as $(x_1, y_1) = (1, 1)$ and $(x_2, y_2) = (2, 2)$.

$$\textcircled{1} \quad a(1) + b(1) + c = a(2) + b(2) + c$$

$$\textcircled{2} \quad a = -b$$

$$\textcircled{3} \quad -bx + by + c = 0$$

$$\textcircled{4} \quad -1 \quad y = x + c, \quad c = 0$$

$\textcircled{5}$ With points, again, lines are parallel,
including $y = x$ and $y = x + 1$.

b) $P^2(F)$ to $P^3(F)$ is a higher-dimensional
projection. Points change from (x, y)
to (x, y, z) . Lines transform between
 $y = ax + b$ and $z = ax + by + c$.

Any two planes in $P^3(F)$ have a
line in common.

Plane and Line in 3D: $ax + by + cz + d = 0$

An example plane intersects point
 $(0, 0)$, with

So, $a(0)+b(0)+cz+d=ax+by+cz+d$
 $y = -\frac{a}{b}x$ is the
common line.

Three noncollinear points lie in
a unique plane:

Three points: $(0,0)$, $(1,1)$, and $(-1,1)$

Equation to a plane in 3D:

$$ax+by+cz+d=0$$

$$\begin{aligned} a(0)+b(0)+cz+d &= a(1)+b(1)+cz+d & (0,1) \\ &= a(-1)+b(1)+cz+d & (-1,1) \end{aligned}$$

$$a = -b, c = -\frac{(2b+d)}{z}$$

Unique plane for three points:

$$-bx+by-2b=0$$

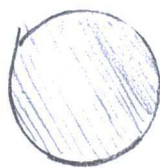
$$-x+y-2=0$$

c) An undefined term in the new space
is the circle.

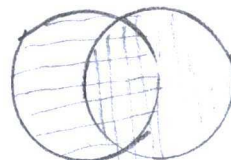
17.

Syllogisms by Lewis Carroll:

a) (from exercise) "No frogs are poetical; but some ducks are unpoetical. Hence some ducks are not frogs."



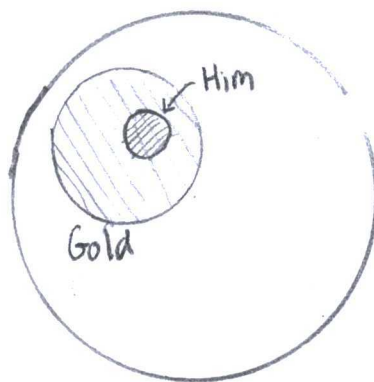
Frogs



Poetical Ducks

True

b) (from exercise) "Gold is heavy; nothing but gold will silence him. Hence, nothing light will silence him."



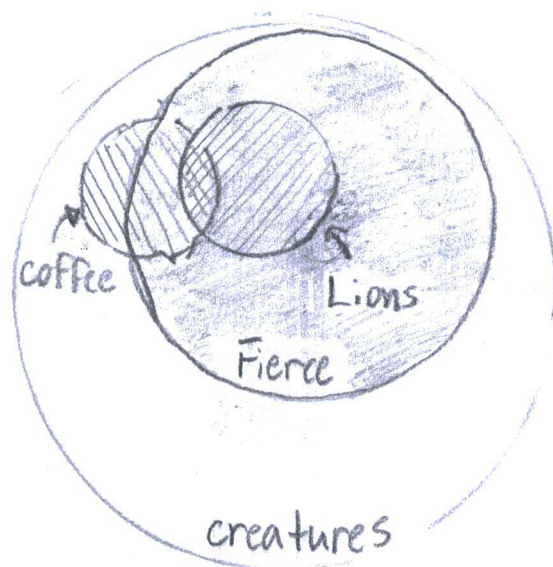
Heavy

True



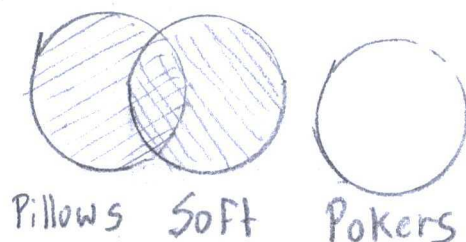
Light

c) (from exercise) "All lions are fierce; some lions do not drink coffee. Hence, some creatures that do not drink coffee are not fierce"



True

d) (from exercise) "Some pillows are soft; no pokers are soft. Hence, some pokers are not pillows."



True

18.

A curved plane has "straight" lines across the surface. Lines transform into Euclidean space from Elliptic or Hyperbolic spaces with a $y=mx+b$ equation.

19. a) Logical Implication: $P \rightarrow Q$ $P = \text{Statement } S$
 $Q = "S \text{ is false}"$

$$P \rightarrow Q = \text{True}$$

$$P \rightarrow \neg Q = \text{False}$$

The self-referential statement is
circular in tautology.

$$b) \exists C \forall S (S \in C \iff S \notin S)$$

$$\text{IF } C = S, \text{ then } C \in C \iff C \notin C$$

The instantiation argues, "If two things
are exact, then the group is a
collection or individuals?"